

Thesis Title: “*Stall-Adaptive Airfoil: Numerical and Experimental Optimization of Dimpled and Leading-Edge Protuberance Surfaces.*”

Roll Numbers: 2002XXX and 2002YYY

Mapping: **Program Outcome (PO1–PO12)**, aligned with the thesis titled “*Stall-Adaptive Airfoil: Numerical and Experimental Optimization of Dimpled and Leading-Edge Protuberance Surfaces*”). It includes:

Code	Attribute / Outcome	Justification in Thesis Context	Reference (Chapter, Section, Page)	Comments
PO1	Engineering Knowledge	Applies aerodynamic theory, biomimetic design, fluid mechanics, and turbulence modeling (DES–SST k– ω).	Chapter 1, Sections 1.1–1.3 (pp. 01–06); Chapter 2, Sections 2.1–2.4 (pp. 08–27); Chapter 3, Section 3.3 (pp. 35–45)	PO1 has been addressed
PO2	Problem Analysis	Identifies aerodynamic stall challenges and analyzes dimple and LEP effects numerically and experimentally across Reynolds numbers.	Chapter 1, Section 1.3 (pp. 04–05); Chapter 4, Sections 4.1–4.6 (pp. 64–112)	PO2 has been addressed
PO3	Design Solutions	Designs stall-adaptive airfoils integrating LEPs and dimples to enhance lift-to-drag ratio and post-stall stability.	Chapter 3, Sections 3.2, 3.4 (pp. 28–32, 43–53); Chapter 4, Sections 4.2 (pp. 66–72); Chapter 5, (pp. 113–116)	PO3 has been addressed
PO4	Investigation	Conducts detailed CFD simulations, optimization studies, Reynolds Numbers scaling, and wind tunnel experiments to validate numerical models.	Chapter 3, Sections 3.3–3.5 (pp. 32–63); Chapter 4, Sections 4.3–4.6 (pp. 72–109);	PO4 has been addressed
PO5	Modern Tool Usage	Utilizes ANSYS Fluent, MATLAB (Response Surface Methodology), and CAD/3D printing tools for analysis and fabrication.	Chapter 3, Sections 3.1–3.5 (pp. 28–63);	PO5 has been addressed
PO6	Engineer and Society	Contributes to safer, efficient aerodynamic systems applicable to UAVs, wind energy, and transportation sectors.	Chapter 1, Section 1.2 (p. 03); Chapter 4, Section 4.7, 4.8.2–4.8.3 (pp. 110–111, 111–112)	PO6 has been addressed
PO7	Environment and Sustainability	Promotes sustainable, energy-efficient aerodynamic designs that reduce drag forces and energy consumption.	Chapter 4, Section 4.8 (pp. 110–112)	PO7 has been addressed

PO8	Ethics	Follows ethical standards in experimentation, data handling, and academic integrity.	Appendix D, (pp. 130-131); References (pp. 117–125)	PO8 has been addressed
PO9	Team Work	Involves teamwork in design, data collection, experimental setup, and validation processes.	Chapter 3, Sections 3.2–3.5 (pp. 29–63)	PO9 has been addressed
PO10	Communication	Presents findings through well-documented thesis, figures, tables, and oral presentations.	All Chapters (pp. 01–115)	PO10 has been addressed
PO11	Project Management	Demonstrates effective planning, budgeting, scheduling (Gantt charts), and resource management.	Appendix, Tables C1–C3 (pp. 129–130)	PO11 has been addressed
PO12	Life-long Learning	Reviews evolving research on biomimetic flow control, LEP/dimple effects, and optimization techniques.	Chapter 2, Sections 2.2–2.4 (pp. 8–27); References (pp. 117–125)	PO12 has been addressed

Knowledge Profile and Complex Engineering Problem

Mapping: *Complex Engineering Problem Attributes* (CEP Attributes P1–P7), aligned with the thesis titled “*Stall-Adaptive Airfoil: Numerical and Experimental Optimization of Dimpled and Leading-Edge Protuberance Surfaces*)”. It includes:

CEP	CEP Attribute	How This Thesis Addresses It	Related Knowledge Profile	References	Comments
P1	Depth of knowledge required	Requires integrated knowledge of fluid dynamics, turbulence modeling (DES/IDDES $k-\omega$ SST), theory of aerodynamics, optimization theory and model, and experimental validation combining fundamentals and advanced biomimetic flow-control theory.	K3: Engineering fundamentals (fluid mechanics, aerodynamics, optimization)	Chapter 2: Sections 2.1–2.3 (pp. 8–15) Chapter 4: Section 4.1 (pp. 64–70)	P1 has been addressed
			K4: Specialist knowledge (turbulence models, optimization, stall control, and vortex visualization)	Chapter 3: Sections 3.3–3.4 (pp. 35–50); Chapter 4: Section 4.3–4.6 (pp. 72–109)	
			K5: Engineering design (custom airfoil geometry with LEPs/dimples)	Chapter 3: Sections 3.2–3.5 (pp. 28–63);	
			K6: Engineering practice (experimental setup and validation)	Chapter 3: Section 3.5 (pp. 50–63); Chapter 4: Section 4.7–4.8 (pp. 110–112)	
			K8: Research literature (biomimetic flow studies and optimization methods)	Chapter 2: Sections 2.1–2.4 (pp. 8–27)	
P2	Range of conflicting requirements	Balances conflicting aerodynamic requirements: maximizing lift while delaying stall; minimizing drag while introducing surface perturbations; optimizing LEP/dimple size while ensuring manufacturability; and maintaining experimental feasibility (blockage limits, low-Re constraints). Performance, cost, and surface complexity trade-offs are analyzed.		Chapter 3: Sections 3.1–3.5 (pp. 28–63); Chapter 4: Sections 4.1–4.6 (pp. 64–109)	P2 has been addressed
P3	Depth of analysis required	Employs extensive CFD parametric studies, compares multiple geometries, multi-objective optimization (RSM / Gradient-based optimization), applies flow visualization		Chapter 3: Sections 3.3–3.5 (pp. 32–63) Chapter 4: Sections 4.2–4.6 (pp. 66–109)	P3 has been addressed

		(vorticity, Q-criterion), conducts mesh independence checks, analyzes Reynolds-dependent behaviour, and correlates CFD predictions with experiments, requiring abstract reasoning and multi-stage analytical development.			
P4	Familiarity of issues	Investigates a new problem area combining LEPs and dimples in stall-adaptive aerofoils, which is seldom explored experimentally in low-Re regimes.		Chapter 2: Sections 2.2–2.3 (pp. 8–26) Chapter 3: Section 3.2 (pp. 29–32)	P4 has been addressed
P5	Extent of applicable codes	The methodology follows accepted aerodynamic testing standards: appropriate tunnel blockage ratios, calibration procedures, repeated measurements, validated CFD practices (mesh quality metrics and y^+ control), proper optimization algorithm and turbulence model selection based on prior benchmark studies.		Chapter 3: Sections 3.3.2–3.3.7, 3.4–3.5 (pp. 33–43, 43–63) Appendix B (pp. 126–128)	P5 has been addressed
P6	Extent of stakeholder involvement and needs	Supports the aerospace and renewable energy industries by developing designs for UAVs and small wind turbines that align with energy-efficiency and safety objectives.		Chapter 1: Section 1.1–1.2 (pp. 1–6) Chapter 4: Section 4.8 (pp. 111–112)	P6 has been addressed
P7	Interdependence	Demonstrates strong coupling between geometry modification, flow separation behaviour, vortex formation, and aerodynamic coefficients (C_L , C_D , L/D). CFD findings and experimental data verify the interconnected nature of these sub-problems, especially around stall and post-stall regimes		Chapter 4: Sections 4.3–4.8 (pp. 72–112)	P7 has been addressed

Mapping: *Complex Engineering Activities*

A1: Uses diverse resources such as equipment, people, software, materials, money, laboratories, and technologies.

Resources	Thesis Components
Software, computational resources	CFD Simulation (ANSYS Fluent)
Software, design resources	SolidWorks
Technologies	3D Printing (Crealty Carbon K1C)
Materials	PLA 3D printing filament
Equipment, Technology	Wind Tunnel Testing
Equipment, People	Data acquisition pitot static tube (velocity, density, pressure and temperature)

A2: Requires resolving significant problems arising from interactions between wide-ranging or conflicting engineering issues.

Significant Problems	Conflicting or Interacting Issues
Need to increase lift without premature stall.	Stall vs. Lift Enhancement
Surface perturbations may improve stall but increase drag.	Drag Penalty vs. LEP/Dimple Geometry
Low-Re aerodynamics are highly unstable and sensitive.	Reynolds Number Sensitivity
Requires reconciling mismatched trends (turbulence, unsteady effects).	CFD vs. Experimental Agreement

A3: Involve creative use of engineering principles and research-based knowledge in novel ways.

Engineering Principles	Novel or Innovative Elements
Novel combination rarely tested in literature.	Hybrid Surface — LEP + Dimples Together
Inspired by humpback whales + golf ball dimples.	Stall-Adaptive Biomimetic Concept
Parametric variation of amplitude, wavelength, and dimple depth.	Geometry Optimization Approach
Innovative turbulence modeling for small-scale airfoils.	Use of DES/IDDES at Low Re

A4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.

Societal/Environmental Impact	Relevance
Improves UAV Safety and Efficiency	Better stall margins → safer operations
Benefits of Small Wind Turbine Airfoils	Improved low-Re performance → higher renewable energy output.
Reduced Energy Waste	Efficient airfoils reduce power loss

A5: Can extend beyond previous experience by applying principles-based approaches.

Unfamiliar Engineering Issues	Relevance
Combined LEP + Dimple Aerodynamic	Very limited previous literature; uncommon design.
Complex Vorticity + Flow Separation Patterns	Requires advanced CFD interpretation (Q-criterion)
Low-Re Stall Physics	Highly nonlinear and difficult to predict